PTERODACTYL v4 carrier — DRAFT generated from tools/generate_v4_schematic.py. Connections by label; see docs/POWER_ARCHITECTURE.md.

Battery -> J1 -> F1 -> Q1 (reverse protection) -> J2 panel switch -> VBAT_SW -> U2 5 V buck-boost -> +5V -> Teensy VIN.

U2 three-pin and U5 GPS carrier footprints assigned. GPS geometry visually accepted 2026-10-20; installed fit, holder clearance and retention pending.

Teensy: cut the VUSB/VIN pads before applying external power (PJRC). +3V3 is the Teensy 3.3 V output (250 mA external max).

I2C at 100 kHz for bring-up (see PLACEMENT_AND_INTERFERENCE.md). No GPIO may exceed 3.3 V.

J1 Battery
F1 3.7A hold
Q1 IRLML6401A
D1 10V
J2 Power switch
C1 100u 25V
R1 100k 1%
R2 22k 1%
C2 100n
U2 Pololu S9V11F5
C3 100u 25V
C4 100n
FB1 75R@100MHz
C5 10u
C6 10u
C7 100n
C8 10u
C9 100n
C10 10u
C11 100n
LED1_A 6824
LED2_A 6825
LED3_A 6826
LED4_A 6827
Teensy 4.1
U3 Adafruit BNO055 #2472
U4 Adafruit MS8607 #4716
U5 Adafruit PA16616D #5440
SW1 Mode
SW2 User
J3 Pull before flight
J4 Spare I/O
D2 Red
D3 Yellow
D4 Green
D5 Orange
TP1 VBAT_PROT
TP2 +5V
TP3 +3V3
TP4 SENSOR_3V3
TP5 GND
TP6 I2C_SDA
TP7 I2C_SCL
TP8 GPS_TXD
TP9 GPS_RXD
TP10 VBAT_SENSE
H1 M3
H2 M3
H3 M3
H4 M3

Generated by tools/generate_v4_schematic.py — not fabrication-approved

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